

An Overview of Fiber-Optic Gyroscopes

(Invited Paper)

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Abstract—The principles of operation of fiber-optic gyroscopes are reviewed. Performance-limiting phenomena are discussed along with methods to reduce their effect on the rotation-rate signal. Current technology and performance of state-of-the-art systems are presented. Finally, a direction of future research and possible applications are indicated.

I. INTRODUCTION

DETECTION of rotation with light was demonstrated by Sagnac in 1913 [1]. He showed that two waves acquired a phase difference by propagating in opposite directions around a loop interferometer (Fig. 1), which was rotating about its axis. The effect was small, however, and measurement of slow rotations required a substantial interferometer. For example, to measure the rotation rate of the earth, Michelson and Gale used an interferometer like the one shown in Fig. 1, but having a perimeter of over 1 mi [2].

A ring laser provides a dramatic improvement in sensitivity, allowing the use of small loops, by internally converting the phase difference to a frequency difference and measuring this frequency difference precisely. Interest in the ring laser gyroscope has continued since its first demonstration by Macek and Davis in 1963 [3]. It has been developed to the point where it is now finding practical application in inertial navigation. While this is an active system, loops which are passive as in Sagnac's case but also resonant as in the ring laser also show promise for such applications, as shown by Ezekiel and Balsamo [4].

An alternative and convenient method for increasing the sensitivity of passive Sagnac systems uses an optical fiber (Fig. 2) which may be wrapped many times around a small cylinder to increase the phase difference produced by rotation. Long total fiber lengths may be used because of the impressive progress which has been made in the fabrication of very low loss single-mode fibers. The first demonstration of such a system was made by Vali and Shorthill in 1976 [5]. This device has since been studied in a number of laboratories, and has progressed from being incapable of detecting the earth rotation rate to being able to detect rotation rates some three orders of magnitude below earth rate today. As a result of this, and also because of the possibility of being packaged in a small rugged

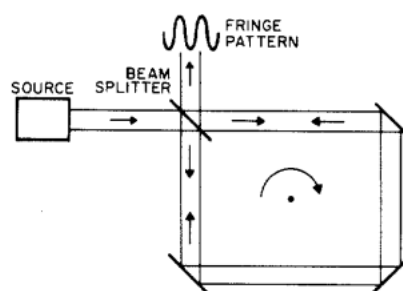


Fig. 1. Bulk-optic loop interferometer.

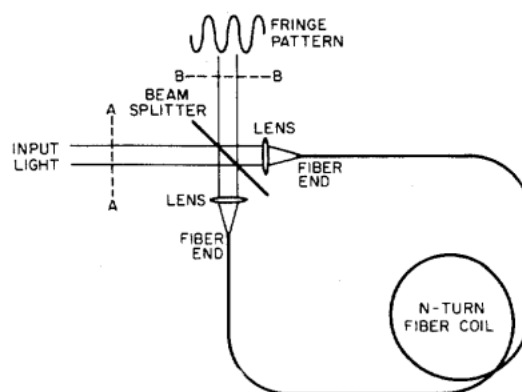


Fig. 2. Loop interferometer with fiber-optic coil.

form, including all-solid-state implementations, fiber gyroscopes are now of substantial interest for potential practical applications.

In the fiber gyro, the insertion of the fiber medium into the optical path of a Sagnac interferometer does not alter the Sagnac effect directly, but gives rise to spurious signals. One problem encountered is associated with birefringence in the fiber when the counterpropagating waves have different states of polarization. Another problem is involved with backscattered waves in the fiber arising from Rayleigh scattering. Both of these phenomena lead to noise, instability, and drift in the signal in response to environmental effects such as vibration and thermal variations. A further problem is concerned with biasing the gyro to move its operating point from one which is insensitive to rotation to one having maximum sensitivity to rotation without, at the same time, introducing further instabilities. Proper consideration for optical reciprocity, optical source characteristics, and biasing requirements allowed fiber gyroscopes to measure below earth rate. To further increase

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the sensitivity to present levels, a number of other physical phenomena had to be identified and accommodated, including the Faraday effect in the fiber material in the presence of the earth's magnetic field and the nonlinear optical Kerr effect in the fiber material. In a general way, the work progressed in cycles consisting of first studying short-term noise over periods of seconds which limited the ultimate rotation sensitivity, followed by attention to long-term drift over periods of hours. As will be seen, there is a current area of research needing further progress, concerned with the stability and dynamic range of the scale factor, which relates the output signal of the device to the rotation rate.

In contrast to conventional gyroscopes, optical gyroscopes have no rotating parts, which, generally, is an asset. They are of interest for many similar applications, including inertial navigation, guidance, and tracking, and applications based on direct determination of the earth's rotation vector. In application, fiber gyroscopes would generally be mounted rigidly on a vehicle or object whose rotation is to be measured, and the only rotation involved is that of the vehicle itself. When used together with accelerometers, the path of a vehicle can be determined entirely from measurements made within the vehicle. Sensitive accelerometers for such purposes are also potentially realizable in a fiber-optic form.

To construct a workable fiber gyroscope a few fundamental elements must be taken into account. These are described in Section II. They have been used to establish the basic design of the most sensitive systems and may serve as a guide to evaluate alternate approaches. Once the fundamental elements are treated, attention must be given to the subtle parasitic effects, which are discussed in Section III. These can usually be removed for the most part with specific operating techniques and component selection, and probably will not require a structural redesign of the system itself. In Section IV we describe the technology of state-of-the-art gyroscopes including current performance levels. The measurement of small rotation rates is very accurate both in the short and long term, but accurate measurement of large rotation rates remains a problem. Section V deals with future research which is largely directed at improving the characteristics of the scale factor and Section VI indicates some possible applications of these systems.

II. FUNDAMENTALS

A. Introduction

This section examines the basic considerations of the fiber gyroscope. We begin by describing the Sagnac effect and by applying it to an interferometer which makes use of a fiber-optic coil (Fig. 2). Then we discuss the principle of reciprocity with which accurate measurements of the rotation rate can be made even if the modal distributions of the two counterpropagating waves are not identical throughout the interferometer. This is applied to the particular case where use is made of a single-mode fiber which guides two polarization modes. We then consider biasing techniques required for precise measurement of small rotation rates. Finally, we look at the performance limitation imposed by photon noise.

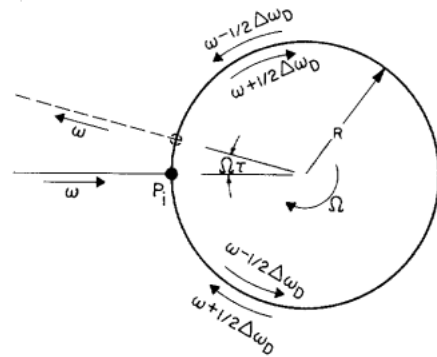


Fig. 3. Schematic of rotating circular loop interferometer showing Doppler shifted frequencies with respect to inertial space.

B. Sagnac Effect

The Sagnac effect has a simple kinematic description [1], [6]. Consider a hypothetical interferometer with a circular optical path in vacuum (see Fig. 3). When the interferometer is at rest in an inertial frame of reference the path lengths of the counterpropagating waves are equal and, since light travels at the same speed c in both directions around the loop, both waves return to the point of injection P_i in phase after a propagation time $\tau = 2\pi R/c$ where R is the path radius. When the interferometer is rotating at a rate Ω and the observer is motionless in the original inertial frame, the point P_i moves through an angle $\Omega\tau$ during the propagation time τ . Therefore, the difference between the propagation times of the counterpropagating waves to the lowest order in $R\Omega/c$ is

$$\Delta\tau = \frac{(2\pi + \Omega\tau)R}{c} - \frac{(2\pi - \Omega\tau)R}{c} = \frac{4\pi R^2\Omega}{c^2}. \quad (1)$$

For continuous waves of frequency ω , this corresponds to a phase shift

$$\Delta\phi = \omega\Delta\tau = \frac{4\pi R^2\omega}{c^2}\Omega. \quad (2)$$

This result remains unchanged when the interferometer is filled with a medium of refractive index n because the Fresnel-Fizeau drag effect due to the movement of the medium compensates for the increased optical path lengths [6]–[8]. A result depending on the material dispersion has been reported [9], but a more detailed analysis taking into account both the Doppler frequency shift due to the motion of the injection point P_i and that, due to the movement of the medium, shows this dependence to be absent. There is no frequency difference between the counterpropagating waves in the rotating frame, the rest frame of the medium, and hence no dependence on dispersion.

In calculating the Sagnac phase difference, we did not consider the Doppler effect. It is not important to do so. Observed from an inertial reference frame, the counterpropagating waves in a rotating vacuum interferometer have different frequencies because of the Doppler frequency shift due to the movement of the injection point moved at a speed of $R\Omega$. The time difference in (1) is unchanged since the speed of light in vacuum is independent of frequency. The frequency in (2) is correct to lowest order in $R\Omega/c$.

The Sagnac phase shift may, however, be calculated using the Doppler frequency shift. In this case we consider the

whole interferometer system at a particular instant. Both optical path lengths are then equal to $2\pi R$, but the propagation constants differ by the Doppler shift $\Delta\beta_D = \Delta\omega_D/c$ (Fig. 3). Therefore at a particular instant in time, the wavelength of the wave propagating in one direction is different from the wavelength of the other wave. The phase shift is then $\Delta\phi = 2\pi R\Delta\beta_D = 2\pi R\Delta\omega_D/c$, with $\Delta\omega_D = 2\omega R\Omega/c$. This leads to $\Delta\phi = 4\pi R^2\omega\Omega/c^2$ which is the same as in (2). There are thus two equivalent interpretations of the Sagnac effect: different propagation lengths leading to different propagation times or different frequencies in two paths leading to different propagation constants with the same optical length.

A more rigorous approach to determine the Sagnac phase difference is to solve electrodynamic propagation equations in the rotating medium [10], [11]. This is most easily accomplished in the rotating reference frame where the boundary conditions are independent of time. The new coordinates of this noninertial frame must be chosen from a relativistic point of view. Electrodynamics laws have a simple form in an inertial frame but can be extended with some modifications to any system of curvilinear coordinates such as, in particular, those of a rotating frame. The main problem is to find the expression for the constitutive tensor which connects the contravariant components of the derived field tensor ($\mathbf{H}$, $\mathbf{D}$) with the covariant components of the macroscopic field tensor ($\mathbf{B}$, $\mathbf{E}$). This relation involves the properties of the medium and noninertial properties of the rotating frame, and it leads to the exact expression for the propagation equation in a rotating frame. This can be solved with a method of perturbation from the solutions for a motionless waveguide-independent inertial frame, which allows one to find the value of the Sagnac phase shift independent of waveguide or medium properties, i.e., index dispersion, phase, and group velocities. Without ambiguity, the phase shift depends only on the light frequency ω and on the dot product of the equivalent area vector of the mean optical path and the rotation rate vector $\mathbf{\Omega}$

$$\Delta\phi = \frac{4\omega}{c^2} \mathbf{A} \cdot \mathbf{\Omega}. \quad (3)$$

It is interesting to note that, for a Sagnac interferometer of arbitrary shape, the phase difference depends on the frequency and not on the wavelength. Thus for acoustic waves rather than light waves the sensitivity would be much lower even with similar wavelengths. The Sagnac effect is due to a difference in the propagation times around a rotating loop, independent of the velocity of the physical phenomenon which is used. The wave frequency serves as a clock to measure this time difference. The higher the frequency, the larger the resulting phase difference.

The advantage of using an optical-fiber coil to form the interferometer is that the Sagnac phase difference increases with the number of turns or the length of the fiber. For this important special case it is often convenient to rewrite the above formulas in terms of the length of the fiber L , the diameter of the coil D , and the vacuum wavelength of the light λ

$$\Delta\phi = 2\pi \frac{LD}{\lambda c} \Omega. \quad (4)$$

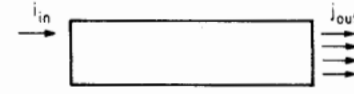


Fig. 4. Schematic of multimode fiber where light enters in a single mode i_{in} and exits through many modes j_{out} .

The increased length that the fiber provides is essential for measurement of rotation rates of interest. An inertial navigation grade system must be able to measure rotation rates of 10^{-2} °/h. The corresponding phase shift, for a wavelength λ of 1 μm , a length L of 1 km, and a coil diameter D of 10 cm, is 10^{-7} rads. This is at the limit of our measurement capability. The optical path length difference is an incredible 10^{-4} Å after the 1-km total path. This is a measurement of 1 part in 10^{17} , so a very high precision is required.

C. Reciprocity

To accurately measure the Sagnac phase difference it is necessary to reduce other phase differences which can vary under the influence of the environment. The degree to which this is possible determines the quality of the gyroscope. This has been the focus of much of the research on these systems to date and is the subject of a major section of this chapter. It is for this purpose that the principle of reciprocity [12] is applied as a foundation upon which the most sensitive systems have been designed. It is used to select portions of the counter-propagating waves which, in effect, pass through the interferometer along a common optical path [13]. Having propagated through the same optical path the two waves have, in the absence of nonreciprocal effects, the same phase delay. Variations of the system by the environment changes the phase of both waves equally so no difference in the phase delay results. In this way, the system obtains a basic degree of immunity to environmental influences that is probably beyond the capability of other phase stabilization techniques applied externally to the interferometer.

To describe the principle of reciprocity, assume an optical system, e.g., a multimode fiber shown schematically in Fig. 4, with certain modal distributions at the input and the output. If light is launched in a mode i_{in} there is, generally, coupling between the different modes during propagation. At the output, light is distributed over the modes j_{out} , with various phases and amplitudes. The important feature of reciprocity of modal propagation is that the phase, which is accumulated by light waves launched at one end of the network in a mode i and coupled into a mode j , is equal to the phase accumulated in the opposite direction by light launched into the opposite end in the mode j and coupled into the mode i . The amplitude coupling coefficients in the two opposite directions are also equal. This relies on the linearity of the propagation equation and the time invariance of the system, as will be discussed later.

A Sagnac interferometer is a special network where the two waves which will propagate through the system in opposite directions enter together before they are divided. In the case of a single-mode input, mode i above is the same as mode j . We need only select that portion of the returning waves which passes back through this same mode to insure that the reciprocal phase accumulated is identical for each of the waves. It is,

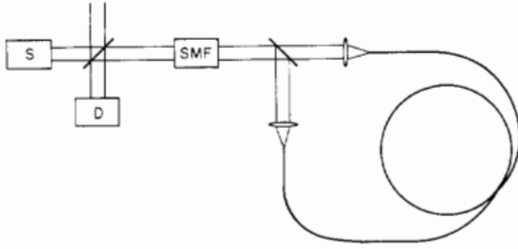


Fig. 5. Loop interferometer with source *S*, detector *D*, and single-mode filter (SMF) at common input-output port for reciprocity.

therefore, sufficient to place a single-mode filter (SMF) at the common input-output port of the interferometer and to detect the waves which pass back through this filter. A schematic of the system is shown in Fig. 5 [14]. A second beam splitter is used to direct some of the returning light to a detector. To maximize the optical power incident on the detector, the optimum splitting ratio of this second beam splitter is 3 dB, which leads to 6 dB to total extra loss because it is passed twice. This loss could be halved with a fast optical switch or eliminated with a Faraday rotator, although this improvement would add complexity to the system. The latter does not cause a reciprocity problem since the Faraday rotator would not be inside the interferometer.

Our ability to make a reciprocal system is partially limited by our ability to make a single-mode filter. A convenient means to do so is with a length of single-mode fiber combined with a polarizer to eliminate one of the two guided polarization modes. The single-mode fiber is an extremely good spatial filter and we assume that it is ideal. However, a finite extinction polarizer may limit the performance of the system. This is treated in the following subsection.

Though implicit in the above statements, we point out explicitly three features of the configuration shown in Fig. 5. First, by using the output port of the interferometer which is the same as the input port, each of the counterpropagating waves undergoes one transmission and one reflection through the beam splitter. If the other port were used, one wave would make two transmissions while the other would make two reflections. The latter would give a spurious phase shift which is tied to the loss in the device [15]. Second, the spatial filter insures that the surviving portions of the two waves take a common path through the beam splitter on transmission as well as on reflection [14]. This is true also for the remainder of the interferometer even if the loop fiber is not a single mode. Third, the polarizer insures that the fiber birefringence will not contribute a phase difference between the two waves which can be very large and subject to change with the environment.

D. Polarization

Cylindrically symmetric single-mode fibers guide two degenerate polarization modes. Real fibers have a small amount of random asymmetry, resulting in a small amount of random birefringence, which varies along their lengths. The state of polarization of the guided wave varies accordingly [16]. In addition, bending and twisting the fiber includes stress which creates additional birefringence [17]–[19]. Therefore, if the states of polarization of the two counterpropagating waves are

not identical their propagation constants are not necessarily the same. Therefore, after passing through the fiber loop, the two interfering waves may not have the same phase. To underscore the importance of this matter, the phase difference due to the bending birefringence alone in a typical fiber coil is likely to be hundreds of radians. This is enormous when one considers that the phase difference due to the Sagnac effect when the gyroscope is rotating at earth rate is on the order of 10^{-4} rad.

A straightforward method to reduce the error is, as mentioned above, to make use of the portions of the waves that have passed through the fiber with the same state of polarization. The state of polarization varies continuously along the length of the fiber. To effect this solution we must use only the portions of the two counterpropagating waves which have identical states of polarization at each point along the fiber. This is insured by reciprocity if a polarizer is used at each end of the fiber [20], [21] or simply with one polarizer at the common input-output port of the interferometer [13], [14]. Then, any phase difference between the two waves which return through the polarizer is not due to the birefringence in the fiber. The amount of light which survives the filtering action of the polarizer does, however, depend on the fiber birefringence and some polarization control may be necessary for this reason.

The quality of the polarizer is a critical factor in the reduction of a phase difference due to birefringence. An ideal polarizer, considered above, removes the phase difference completely by reciprocity. Unfortunately, real polarizers do not entirely eliminate the cross-polarized wave and this added to the signal wave alters its phase. A term proportional to the amplitude extinction ratio of the polarizer can contribute a significant phase difference even for very high quality polarizers [22]. The maximum phase error to first order in the amplitude extinction ratio ϵ is

$$\phi_{\epsilon \max} = \epsilon \frac{2|a_B||t_{AB}|}{|a_A||t_{AA}|} \quad (5)$$

where a_A and a_B are the amplitudes of the source light incident on the principal axes of the polarizer. The polarizer passes a_A efficiently and attenuates a_B by the amplitude extinction ratio ϵ . The quantity $|t_{AB}| = |t_{BA}|$ is the fraction of ϵa_B that is returned through the fiber loop to the *A* principal axis of the polarizer. The fraction of a_A that is returned to this axis is $|t_{AA}|$. If the two waves $\epsilon a_B t_{AB}$ and $a_A t_{AA}$ at the output of the polarizer are in phase quadrature, their interference creates the maximum phase error given above. Reduction of ϵ , $|a_B|$ or $|t_{AB}|$ will reduce the maximum possible error. The error is thus determined by the quality of the polarizer and by the control of the state of polarization. By requiring that $\epsilon \leq 10^{-4}$, $|a_B|/|a_A| \leq 10^{-1}$, $|t_{AB}|/|t_{AA}| \leq 10^{-1}$, the phase error is $\phi_{\epsilon} \leq 10^{-6}$ which is acceptable only for medium sensitivity applications. Thus very high quality polarizers are required as well as accurate polarization control.

An alternative approach makes use of unpolarized light without the polarizer in the system [23]. The two polarization modes of the fiber in the interferometer are then excited equally and the error from one serves to cancel the error from the other.

The problem is now to reduce the degree of polarization to a level where this cancellation is nearly complete. The requirements are stringent and not likely to be met. Therefore, modifications to the system may be necessary in order for the use of unpolarized light to become a practical alternative.

The restrictions in both the polarized and the unpolarized systems may be reduced with the use of highly birefringent polarization preserving fiber and a wide-band source [24]. With this combination the intensity of the portions of the waves which produce the error may be reduced due to the reduction in their mutual coherence.

E. Bias

The measured optical power from the interferometer is proportional to the square of the absolute value of the detected electric field. The electric field is the sum of the electric fields of those portions of the counterpropagating waves which reach the detector. Writing the detected power P_D in terms of the optical powers of the surviving portions of the interfering waves P_1 and P_2 we have

$$P_D = P_1 + P_2 + 2\sqrt{P_1 P_2} \cos(\phi_1 - \phi_2) \quad (6)$$

where ϕ_1 and ϕ_2 are the phases of waves 1 and 2, respectively, incident on the detector. In a reciprocal system the powers and phases of these waves are equal. In general we may express these quantities as

$$\begin{aligned} P_1 &= \frac{1}{4} P_0 + \frac{1}{2} \Delta P_{NR} & P_2 &= \frac{1}{4} P_0 - \frac{1}{2} \Delta P_{NR} \\ \phi_1 &= \phi_0 + \frac{1}{2} \Delta \phi_{NR} & \phi_2 &= \phi_0 - \frac{1}{2} \Delta \phi_{NR} \end{aligned} \quad (7)$$

where P_0 is, without loss, the input power and where ΔP_{NR} and $\Delta \phi_{NR}$ are the nonreciprocal power and phase differences, respectively. Combining (6) and (7) we have

$$P_D = \frac{1}{2} P_0 + \frac{1}{2} P_0 \left(1 - 2 \frac{\Delta P_{NR}^2}{P_0^2} \right) \cos \Delta \phi_{NR} \quad (8)$$

to the second order in $\Delta P_{NR}/P_0$. Since $\phi_0 \sim 10^{10}$ around a 1-km fiber loop, a very small fractional phase difference $\Delta \phi_{NR}/\phi_0$, i.e., on the order of 10^{-10} , can result in significant phase difference. A similar fractional power difference will leave the system relatively unchanged. Neglecting the power difference, in the presence of an assumed small nonreciprocity, the detected power is

$$P_D = \frac{1}{2} P_0 (1 + \cos \Delta \phi_{NR}). \quad (9)$$

If the total nonreciprocal phase difference is much less than 1 rad, the cosine will be near its maximum value and the detected power will not be sensitive to the small phase shifts due to rotation. It is, therefore, important to add a biasing phase difference to shift the signal, avoiding both the maximums and minimums of the sinusoid. The problem with doing so is that the biasing phase difference must be stable, or at least known, to a very high precision, so as not to obscure the phase difference due to the Sagnac effect.

A biasing phase difference $\Delta \phi_b$ can be used to shift the operating point to the slope of the cosine as shown in Fig. 6(a). A variety of techniques can be used for this purpose including a rotation, a magnetic field, or a frequency difference between the counterpropagating waves through part of

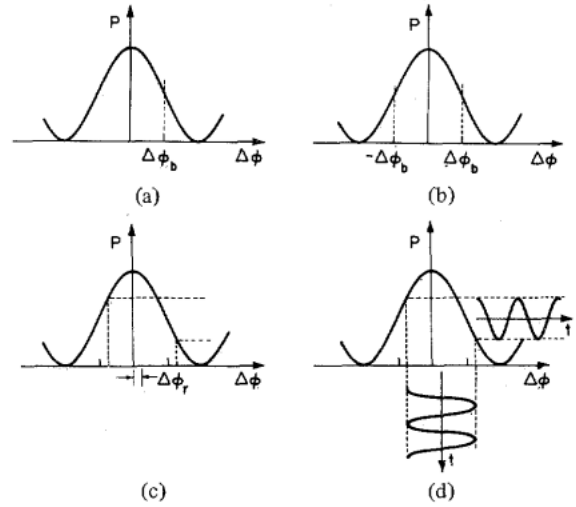


Fig. 6. Optical power versus phase difference between interfering waves with (a) phase bias at $\Delta \phi_b$, (b) symmetrical phase bias at $\pm \Delta \phi_b$, (c) symmetrical phase bias including rotation phase difference $\Delta \phi_r$, and (d) dynamic phase bias as a function of time with rotation and showing bandpass filtered signal power as a function of time.

the loop. The detected power is then proportional to small rotation rates. Unfortunately, changes in the detected power due to variations in the system including a variation in the source power can be confused with rotation. To remove this problem an equal negative phase difference $-\Delta \phi_b$, may be applied as in Fig. 6(a). The power difference between the two biased signals is zero without rotation, independent of the amount of light reaching the detector. With small rotation the power difference varies linearly (see Fig. 6(c)). A two-polarization gyroscope, biasing one state of polarization at $\Delta \phi_b$ and the other at $-\Delta \phi_b$ by using a Faraday cell, was demonstrated [25]. The most successful biasing technique makes use of an alternating phase bias, as shown in Fig. 6(d) [14], [26]. The amplitude of the optical power which varies at the frequency of phase modulation is then proportional to small rotation rates. By biasing the signal on a high-frequency carrier, substantial electronic $1/f$ noise is removed and a phase sensitive detection system can be used.

To create the phase difference modulation, a phase modulator is placed at one end of the fiber loop (see Fig. 7) [14], [26]. The two counterpropagating waves receive the same phase modulation but at different times. The time difference is τ' , the group delay time between the phase modulator and the symmetrical point on the other side of the fiber loop. This results in a phase difference modulation between the interfering waves given by

$$\Delta \phi(t) = \Delta \phi_{NR} + \phi_m(t) - \phi_m(t - \tau') \quad (10)$$

where $\phi_m(t) = \phi_{m0} \sin(\omega_m t)$ is the applied sinusoidal modulation and ω_m is the frequency of modulation. The component of the detected power which varies at the modulation frequency is [14]

$$P_D(t) = 2P_0 J_1 \left(2\phi_{m0} \sin \frac{\omega_m \tau'}{2} \right) \sin \Delta \phi_{NR} \sin \omega_m t \quad (11)$$

where $J_1(2\phi_{m0} \sin \omega_m \tau'/2)$ is the first-order Bessel function of the first kind which takes its maximum value at $2\phi_{m0}$

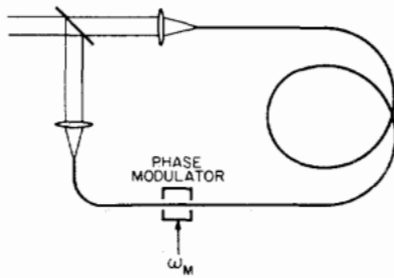


Fig. 7. Loop interferometer with a biasing phase modulator.

$\sin \omega_m \tau' / 2 = 1.8$. Note that as the nonreciprocal phase difference approaches zero the signal disappears. The magnitude of the signal at this frequency is determined by a phase sensitive detector and reveals the nonreciprocal phase difference. In the absence of other nonreciprocal phase differences, it reveals the rotation rate. In principle, almost any modulation frequency is adequate. However, spurious signals which arise due to an imperfect phase modulator may be eliminated by operating at $\omega_m = \pi/\tau'$ [27]. As a numerical example, the proper modulation frequency is approximately 100 kHz for a 1-km fiber loop.

F. Photon Noise

The fundamental limit on the measurement capability of fiber gyroscopes is set, as in any interferometer, by photon shot noise. The amount of noise depends on the optical power incident on the detector. To estimate the magnitude of the noise we assume that the system is statically biased so that the power dependence on the Sagnac phase shift is converted from (9) to

$$P_D = \frac{1}{2} P_0 (1 + \sin \Delta\phi_r) \quad (12)$$

for sensitive measurement of the rotation rate. Here the shot noise due to the detected power is confused with phase noise and causes the rotation-rate error.

Photon shot noise is the result of the quantum nature of light. The number of detected photons per unit time is not constant, but rather varies randomly with a Poisson probability distribution. The average number of photons incident on the detector within an integration time T is simply the average incident energy $(1/2) P_0 T$, divided by the energy of a single photon $h\nu$

$$\bar{N} = \frac{1}{2} \frac{P_0 T}{h\nu} \quad (13)$$

where h is Planck's constant and ν is the optical frequency in cycles per second. The standard deviation of the number of photons is, according to Poisson statistics, equal to the square root of the mean value

$$\sigma = \sqrt{\bar{N}}. \quad (14)$$

The corresponding rms value of the phase noise is given from (12) and (14) by

$$\Delta\phi_{rms} = \frac{P_{rms}}{P_0} = \frac{\sigma}{\bar{N}} = \frac{1}{\sqrt{\bar{N}}}. \quad (15)$$

Using (13) we have

$$\Delta\phi_{rms} = \sqrt{\frac{h\nu}{\frac{1}{2} P_0}} B \quad (16)$$

where $B \equiv 1/T$ is the bandwidth of the detection system. As a numerical example we assume $\frac{1}{2} P_0 \sim 100 \mu\text{W}$ and $\nu = 3 \times 10^{14}$ Hz corresponding to a vacuum wavelength of $1 \mu\text{m}$

$$\frac{\Delta\phi_{rms}}{\sqrt{B}} \sim \frac{10^{-7}}{\sqrt{\text{Hz}}}. \quad (17)$$

To determine an expression for the rms rotation-rate error Ω_{rms} due to photon noise, we use (16) in (4) and obtain

$$\frac{\Omega_{rms}}{\sqrt{B}} = \frac{c^2}{2\pi v L D} \sqrt{\frac{h\nu}{\frac{1}{2} P_0}} \quad (18)$$

where we have let $\lambda = c/\nu$.

We see that the ultimate measurement capability of the gyroscope is strongly affected by the dimensions of the fiber coil. Noting that P_0 is a function of the fiber length and attenuation, an optimum value of the length can be determined [5], [28]. For state-of-the-art low-loss fibers, the optimal length is many kilometers. In practice, however, a typical system has a fiber length of 1 km and a coil diameter of 10 cm. This, together with the values chosen above for the optical frequency and power, gives

$$\frac{\Omega_{rms}}{\sqrt{B}} \sim 10^{-2} (^\circ/\text{h})/\sqrt{\text{Hz}}. \quad (19)$$

With the bandwidth expressed in the commonly used units of inverse hours we have

$$\frac{\Omega_{rms}}{\sqrt{B}} \sim 10^{-4} ^\circ/\sqrt{\text{h}}. \quad (20)$$

This photon noise level is, to a good approximation, unchanged in the case of the dynamically biased system discussed above. It is not restrictive in most applications. Commercial inertial navigation can tolerate noise levels at an order of magnitude larger than this value. In addition, many other applications are even more tolerant. For applications which require an extremely high precision, but which can tolerate a larger system, the quantity LD can be increased by one or two orders of magnitude or more. This provides a corresponding reduction in the rms rotation-rate error.

III. PARASITIC EFFECTS

A. Introduction

A variety of parasitic effects can create phase differences in the fiber gyroscope comparable to, or greater than, the Sagnac phase difference for the earth's rate of rotation, i.e., $15 ^\circ/\text{h}$. Their identification and methods to reduce the resulting errors have been the focus of much research during the past few years. We discuss errors resulting from environmentally induced fluctuations in fiber parameters and offsets due to Rayleigh scattering, the Faraday effect, and the optical Kerr

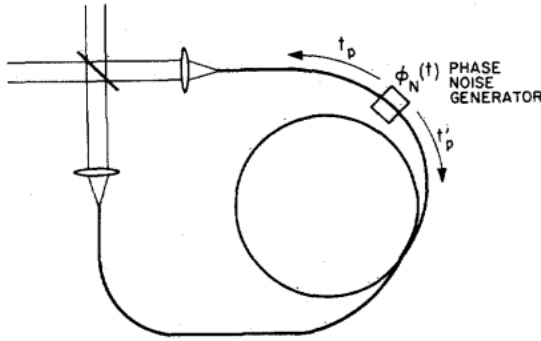


Fig. 8. Loop interferometer with phase-noise generator at an arbitrary position for analysis.

B. Direct Dynamic Effects

Random time variations in ambient temperature and in mechanical stress exerted on a fiber can lead to changes in the optical propagation constants and the fiber dimensions, imparting phase noise to waves passing through the fiber. This can result in a phase difference noise $\Delta\phi$ at the detector, since reciprocity strictly applies only to time invariant systems [29], [30]. Consider a single phase noise source located at an arbitrary point along the fiber loop (Fig. 8), which imparts a random phase increment, having spectral density $\gamma_\phi(\omega)$, to each of the two counterpropagating waves. The spectral density of $\Delta\phi(t)$ is then [30]

$$\gamma_{\Delta\phi}(\omega) = 2\gamma_\phi(\omega)[1 - \cos(\omega\Delta t)] \quad (21)$$

where $\Delta t = t_p - t'_p$ is the difference between the propagation times in the two opposite directions between the noise source and the loop directional coupler. For low frequencies $\omega \ll \pi/\tau$, where τ is the group transit time around the fiber loop, $\gamma_{\Delta\phi}(\omega) \simeq \omega^2 \Delta t^2 \gamma_\phi(\omega)$. It is evident that phase noise sources near the ends of the loop, where Δt is largest, are the most important. Furthermore the highest frequency components within the passband of the detection system are the most heavily weighted. As an example, for a 1-Hz phase noise source at the end of a 1-km fiber loop, $\gamma_{\Delta\phi}$ is nine orders of magnitude smaller than γ_ϕ .

From the above it is clear that, by the nature of the reciprocal closed-loop interferometer, the effect of environmentally induced phase noise is considerably reduced. Further reduction is possible by considering the spacial distribution of phase noise sources. For an arbitrary distribution an integration as a function of distance along the loop is needed to determine $\Delta\phi$. It is obvious, however, that a distribution which is symmetrical about the center point of the loop produces zero net $\Delta\phi(t)$. This situation can be approximated by wrapping the fiber such that points which are equally spaced along the fiber from the center point of the loop lie in close mutual proximity [27], [29].

C. Rayleigh Backscatter

Radiation backscattered from the primary counterpropagating waves in a fiber gyroscope gives rise to a secondary pair of waves guided by the fiber which can be a source of large random errors [31], [32]. These errors can be reduced to a satisfactory level by introducing the following technique.

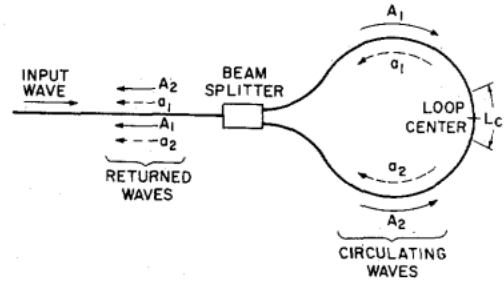


Fig. 9. Loop interferometer with primary wave amplitudes A_1 and A_2 and secondary backscattered wave amplitudes a_1 and a_2 . The backscattered waves which are coherent with the primary waves are scattered from within a coherence length L_c of the center of the loop.

light and by using heterodyne detection involving modulation applied within the sensing loop.

Portions of Rayleigh backscattered radiation and of reflections from discontinuities such as splices, are captured and guided by the fiber, giving rise to secondary waves a_1 and a_2 , which are derived from primary waves A_1 and A_2 , respectively (Fig. 9). As the four waves emerge from the loop, it is perhaps convenient to think of them as being paired such that the portion of the amplitude of wave a_2 , which is coherent with that of wave A_1 , adds vectorially thereto, and similarly for waves a_1 and A_2 . This, in general, alters the phases of the original two waves by different amounts, because waves a_1 and a_2 do not satisfy reciprocity as the primary waves do. This results in an error in the phase difference $\Delta\phi$, which varies with the environment due to the different optical paths, leading to a time-varying error in the rotation signal. This causes noise in the detected signal since an appreciable amount of the time-varying error remains within the passband of the detection system, typically 0-1 Hz.

Several techniques for reducing the phase error by averaging it with the detection integration time have been demonstrated. Phase modulation applied at the input before the fiber loop [33], at one end of the loop [32], and at the center of the fiber loop [34] proved to be useful when using a relatively coherent source. Low coherence source radiation has, however, been the most successful. This has been accomplished with a laser diode [30], a super radiant diode [35], or a gas laser with a time-varying cavity length [36]. Only the scattered light which is partially coherent with the primary signal wave contributes a nonnegligible slowly varying phase perturbation. Therefore, it is the light scattered from the section of fiber within a coherence length of the center of the fiber loop which is of the greatest concern. To make an estimate of the required coherence length we note that scattering from a 1-km fiber loop can cause a phase error on the order of 10^{-2} rad [31] which corresponds, for a 10-cm-diam coil, to a rotation rate error of 1000 °/h. For inertial navigation requirements, this error must be reduced by more than five orders of magnitude. Since it is the amplitude, rather than the intensity, of the scattered wave which is relevant to reduce the phase error by five orders of magnitude, the length of the contributing section of fiber must be reduced by ten orders of magnitude. This would require a coherence length of less than 1 μm , or less than a wavelength. This is impractical, and, fortunately, the error can be reduced by more than five orders of magnitude by using heterodyne detection involving modulation applied within the sensing loop.

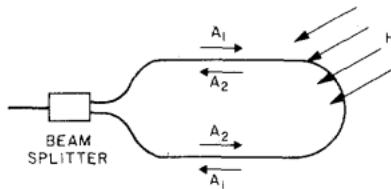


Fig. 10. Loop interferometer in a uniform magnetic field.

For gyroscopes biased with a reciprocal phase modulator at one end of the fiber loop (Section II-E), it has been shown that with the proper choice of the modulation frequency, the light scattered from the center of the loop does not contribute to the output signal at the modulation frequency [37]. Therefore, this modulation technique provides not only a stable bias but also isolation between the signal and the noise due to coherent backscattered waves.

D. Faraday Effect

A uniform magnetic field can, via the Faraday effect, cause a rotation-rate error in the fiber gyroscope [29], [38]–[40]. This error is an offset which depends on the magnitude and orientation of the field as well as the fiber birefringence throughout the interferometer. A change in any of these quantities will cause a corresponding change in the offset. This often appears as a drift in the rotation rate. The error due to the earth's magnetic field is typically $10^\circ/\text{h}$. To date, the problem has been treated by shielding the interferometer from the magnetic fields, though control of the states of polarization of the waves to reduce the sensitivity of the system may also be effective.

The effect of the magnetic field is to induce a nonreciprocal circular birefringence, i.e., birefringence that depends upon the direction of propagation, which adds to the already present reciprocal birefringence in the fiber. The combination of the two causes the system to be sensitive to environmental magnetic fields. Without the reciprocal birefringence the nonreciprocal phase difference would integrate to zero around the closed fiber loop because the line integral around the loop of the tangential component of an environmental magnetic field is zero. Referring to Fig. 10, the nonreciprocal phase accumulated on one side of the fiber loop would be cancelled by the nonreciprocal phase of the opposite sign accumulated on the other side of the loop when the waves reverse their directions of propagation with respect to the field. When reciprocal birefringence is included, this cancellation is not complete.

The birefringence in the fiber coil is not optimized for magnetic field detection and some cancellation exists. Several groups have independently found a reduction of 10^{-3} from the predicted value for the error due to the earth's magnetic field, but the offset remained large at $10^\circ/\text{h}$ [29], [38]–[41]. It changed, as expected, with the orientation of the loop, and was reduced with a magnetic shield in each case.

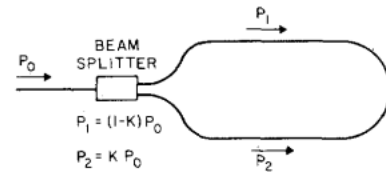


Fig. 11. Loop interferometer with optical powers of counterpropagating waves.

E. Optical Kerr Effect

The optical Kerr effect is a third-order nonlinearity which manifests itself as an intensity-dependent perturbation of the refractive index of a material. In a single-mode fiber this means that the propagation constant of the guided wave becomes a function of the power of that wave. This results in a rotation-rate error in the gyroscope if the optical powers, and, therefore, the propagation constants, of the two counterpropagating waves are not the same [42], [43]. Surprisingly, a power difference as small as 10 nW in as linear a material as fused silica creates an error which is intolerable for inertial navigation. Variation in this power difference can result from changes in the environment, causing drift in the signal away from the proper measure of the rotation rate. A total power of about 100 μW , or more, is needed in typical cases in order for the output signal at low rotation rates to override electronic or photon noise. Therefore, the power difference may need to be controlled, or known, to better than one part in 10^4 of the total power. This would be a difficult task. Fortunately, the tolerance can be increased to a practical value by modulating the source power [44], [45] or by using one with suitable spectral characteristics [45]–[48], as described below.

The propagation constants β_1 and β_2 of waves 1 and 2 are modified by addition of power dependent increments β_{k1} and β_{k2} given by

$$\begin{aligned}\beta_{k1} &= \gamma[P_1 + 2P_2] \\ \beta_{k2} &= \gamma[P_2 + 2P_1]\end{aligned}\quad (22)$$

where P_1 and P_2 are the optical powers of the waves (Fig. 11) and γ is a constant. The power of wave 2 affects the propagation constant of wave 1 twice as much as the power of wave 1. Similarly, for wave 2, the cross effect is twice the self effect. This causes the propagation constants to be different if the powers of the two waves are not identical. If the perturbations were to depend simply on the sum of the two powers, this would not be the case.

If the source power, $P_0(t)$ in Fig. 11, is relatively constant for periods which are long in comparison with the transit time around the loop and if we neglect the attenuation in the fiber, the phase difference due to the Kerr effect and the resulting rotation-rate error are easily determined. If, on the other hand, the source power varies significantly during a transit time, the calculations become more complex. The result is, in any case, simple if we assume that the average value of the source power is established in any arbitrary time increment equal to twice the loop transit time. Additionally assuming that the system

is suitably biased we find that the rotation-rate error Ω_k is given by [44]

$$\Omega_k = \gamma'(1 - 2K) \left[\frac{\langle P_0^2(t) \rangle - 2\langle P_0(t) \rangle^2}{\langle P_0(t) \rangle} \right] \quad (23)$$

where γ' is a constant, K is the splitting ratio of the loop coupler, and the angled brackets indicate a time average. A similar power dependence is obtained in a more general calculation which takes into consideration arbitrary elliptical states of polarization and attenuation in the fiber [45].

A duty cycle $\langle P_0(t) \rangle^2 / \langle P_0^2(t) \rangle$ of $\frac{1}{2}$, or a coupling ratio K of $\frac{1}{2}$, will eliminate the rotation rate error due to the Kerr effect. The tolerance on either, used independently, is of the order of $\pm 10^{-4}$ for long-range inertial navigation applications. They may, however, be used together to meet the requirements with more reasonable tolerances, e.g., $K = \frac{1}{2} \pm 10^{-2}$ and $\langle P_0(t) \rangle^2 / \langle P_0^2(t) \rangle = 1/2 \pm 10^{-2}$.

Though a modulation may be applied to the source, a 50-percent duty cycle occurs naturally in the power from a polarized thermal source, or, similarly, from a superradiant diode [45]–[48]. In addition many multimode sources also tend to cancel the Kerr-induced rotation-rate error without externally applied modulation [45], [46], [48]. This characteristic makes these sources attractive for application in the fiber gyroscope.

IV. TECHNOLOGY

A. Introduction

The first fiber gyroscope used a single-mode fiber for the sensing loop, bulk-optic components to manipulate the light, and a helium-neon laser source. Since that time diode sources and guided-wave components, designed in part for this application, have been used in a number of systems. Alternate types of single-mode fiber, including highly birefringent fiber and polarizing fiber, are being studied for this application. In this section we review the status of fiber and integrated optic components, the sources and the optical fibers in connection with their roles in fiber gyroscopes. We will discuss the effects of the various types of devices on the performance of the most sensitive systems. We will see that tremendous strides have been made in reducing both short- and long-term noise levels and that, as a result, the fiber gyroscope shows promise for use in a wide variety of systems.

B. Optical Circuits

It is advantageous to determine the simplest optical circuit for the fiber gyroscope which will satisfy the operating requirements. These requirements are determined by the application. A circuit that is fine for high-accuracy applications may be unnecessarily complicated for low-accuracy applications. The possible applications extend from long-range navigation requiring measurement accuracies of less than 10^{-2} °/h to short-term guidance where required accuracies may be tens of degrees/hour. Therefore, it is likely to be important to tailor the system design to the requirements. As it turns out, the systems

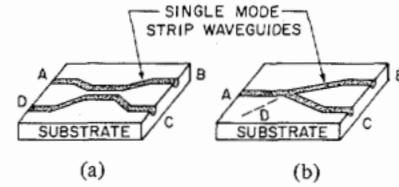


Fig. 12. Integrated-optic four-port directional couplers where the fourth port D and the Y-junction are a radiation mode in the substrate. (a) Evanescent field coupler. (b) Y-junction.

which have demonstrated the highest accuracy have been among the simplest. The reason is that once the gross errors are eliminated with the reciprocal configuration combined with the dynamic bias, the parasitic signals can be removed with the proper types of components and operating techniques and often without additional components. The technology involved in this basic system is the major thrust of this section.

It is important to note that there is the possibility of further simplification of the above mentioned system. The reciprocal configuration (see Fig. 5) used in single-mode fiber gyroscopes to reduce the rotation-rate error due to the environmentally dependent fiber, birefringence, and environmentally dependent splitter/combiner parameters is discussed in Section II-C and II-D. It is possible that the fiber birefringence may be dealt with in other ways, e.g., by using single polarization fiber [49] or by using unpolarized wide-band sources [23], [35], possibly in combination with highly birefringent polarization preserving fiber [50]. This could remove the need for a polarizer or at least reduce its required quality [24]. It is also possible that the error due to the splitter/combiner [51] could be made small and stable enough so as to be inconsequential for low-sensitivity applications. This together with the alternative solution for the birefringence would allow for the use of the nonreciprocal port of the interferometer, removing even the need for the second power splitter.

C. Integrated-Optic Components

An integrated-optics approach to the components offers the possibility of mass production and, therefore, low cost. It is possible that the necessary functions for a three-axis system can all be performed on a single chip. Such systems were proposed early [52], [53], but problems with the devices and with the interfaces between the chip and the fiber have delayed their fruition.

Two types of guided-wave power splitter/combiners have been demonstrated. One involves bringing two waveguides side by side, as in Fig. 12(a), allowing for a power transfer via the evanescent fields of their modes [54]–[56]. The splitting ratio is critically dependent on the device parameters and usually requires adjustment while monitoring the outputs of the device. Since the relative positions of the guides are determined before their fabrication, geometrical adjustment is not possible. The alternative is to adjust the difference between the propagation constants of the waves in the two guides. This is accomplished by making the guides in an electrooptic substrate and depositing properly positioned electrodes for appli-

cation of control fields. The second type of splitter/combiner is a Y-junction in which one guide branches into two as in Fig. 12(b) [57], [58]. In integrated optics this device has the distinct advantage that its splitting ratio can be determined prior to the fabrication process by its symmetry, and no control is necessary after the waveguides are formed. Propagating from left to right, light inserted into port *A* is equally divided between ports *B* and *C*. In the reverse direction, however, some light is generally lost from the guiding structure into a fourth port *D* by radiating into the substrate as indicated by the dashed line in the figure [57]–[59]. Thus the Y-junction has, in effect, four ports just as does the evanescent field coupler. Power lost to the radiation port *D* for the Y-junction does not represent a disadvantage with respect to the evanescent field coupler for most gyro applications, because the same power is delivered to port *D* in both devices, and is generally discarded. In both devices, the amount of power returned by port *A* when propagating from right to left depends on the relative amplitudes and phases of the light inserted into ports *B* and *C*. This is all that is necessary to sense rotation.

Integrated-optics polarizers exploit the differences between the quasi-TE and -TM guided modes in order to, ideally, eliminate one without attenuating the other. Metallic films deposited on the surface of substrates containing waveguides have received a substantial amount of attention for this purpose [60]–[62]. Even so, reported extinction ratios have been limited to about 20 dB. Devices using semiconductor films [63] have also been demonstrated, but so far without more success. Alternatively, bulk dielectric material may be used instead of films to couple the unwanted polarization from the waveguide [64], [65]. This type of polarizer has the potential of a very high extinction ratio as has been realized in its fiber-optic counterpart discussed below. Still another candidate, the reduced-depth polarizer, makes use of precisely controlled waveguide dimensions to cause one of the polarization modes to be cut off [66]. The extinction ratios in this type of device have also been less than 20 dB.

Integrated-optics phase modulators are made by fabricating waveguides in an electro-optic substrate, e.g., LiNbO_3 , and applying electric fields via deposited electrodes [67]. Though the use of this type of device is more complicated than the analogous fiber optic phase modulators, it has the advantage of high-frequency large-bandwidth operation [67].

D. Fiber-Optic Components

Fiber-optic components manipulate the light while it remains within the optical fiber itself. Their use in the fiber gyroscope places the sensing loop and the components on common ground. The components may be attached to the sensing loop with established splicing techniques or fabricated directly on the same fiber without splices. Either method results in little loss, e.g., 0.5 dB/splice or less, and virtually no reflected light. The components themselves can be made to have low loss, high stability, and high directivity.

Phase modulation is induced mechanically, often with a short section of fiber attached to a piezoelectric ceramic. The applied time-varying stress modulates the fiber length and refractive indexes and, therefore, the phase of the guided

wave. This may be accomplished without even interrupting the fiber jacket by wrapping the fiber around a piezoelectric tube [68] whose radius expands and contracts with an appropriate electrical signal. Due to the simplicity of the device, it has been used in many systems. As alternatives, progress is being made on piezoelectric and piezomagnetic fiber jackets [69].

Fiber-optic polarization controllers adjust the state of polarization of the guided wave by manipulating birefringence either intrinsic to the fiber [70] or stress induced by squeezing [71] or bending [18], [19] the fiber. One simple device consists of two fiber loops [19] to induce bending birefringence whose principle axes are in the plane of each loop and perpendicular to it. Any desired transformation of the state of polarization from one side of the device to the other is achieved by adjusting the orientations of the planes of each of the loops. As with the phase modulator, this polarization controller can be easily attached to the fiber loop without splicing and without interrupting the fiber jacket.

The construction of fiber-optic directional couplers requires access to the evanescent fields outside the cores of two fibers. This has been accomplished by removing parts of the cladding of each fiber by chemical etching [72] or grinding and polishing [73]–[75] techniques. Fusing two fibers together while stretching them to enlarge their modes for significant overlap has also been successful [76]. In addition, fibers specially prepared with nonconcentric cores near the surface of the cladding can be positioned for coupling with only the removal of a plastic jacket [77]. All these procedures are delicate and require high precision. As a result, although their performance is excellent, directional couplers have thus far been used in only a few gyroscopes. It is important to note that both fused [78], [79] and polished [80], [81] couplers have been successfully fabricated using highly birefringent fiber and that at least one such device has been incorporated into a gyroscope [50].

Fiber-optic polarizers have made use of mechanisms first described in connection with the integrated optic polarizers mentioned above. They have, however, been somewhat more successful. The metal-film approach has yielded extinction ratios up to about 40 dB [82], [83]. Coupling the unwanted polarization from the guide into a bulk material, i.e., a properly oriented birefringent crystal of potassium pentaborate ($\text{KB}_5\text{O}_8 \cdot 4\text{H}_2\text{O}$), has provided extinction ratios of 90 dB [30], [84]. A recently proposed alternative is to use a fiber designed to guide only one polarization mode [49]. This might eliminate the problem of the complicated fabrication procedure.

E. Sources

There are a few constraints on the source for a fiber gyroscope. Probably the most obvious is that it must be able to inject a significant amount of optical power, e.g., 100 μW or more, into a single-mode waveguide. This condition is met with the proper focusing of light from most lasers which oscillate predominantly in a single transverse mode. Nearly all of the light, aside from reflections, can be coupled from a He-Ne laser into a single-mode fiber with one or more spherical lenses since the transverse distributions of the light from

both are approximately circular gaussians. The light from a single-transverse-mode diode laser is not as easily inserted since it typically has an elliptical transverse distribution and is not without aberration. Even so, coupling efficiencies between 10 and 20 percent are often obtained by butting the fiber against the output face of the laser, and efficiencies up to 50 percent are obtained with lenses [85], [86]. Slightly reduced, but still adequate, coupling efficiencies are obtained with superluminescent diodes [87] which have been known to inject 500 μ W into a single-mode fiber [88]. Due to a very low coupling efficiency, thermal sources are unlikely candidates for use in the fiber gyroscope.

Another restriction on the source is set by noise due to Rayleigh backscattered light in the fiber [31], [32], [45]. This has been reduced primarily by reducing the coherence length of the source light (see Section III-C). To achieve current state-of-the-art short-term levels, it appears that the source-coherence length must be less than, or approximately, 1 cm [45]. Some diode sources naturally have coherence lengths of less than 1 mm, and, therefore, satisfy this constraint without modification. To reduce the coherence length of other lasers, such as an He-Ne laser, the cavity modes can be swept across the gain curve of the amplifying medium. This can be accomplished by dithering one of the cavity mirrors with, for example, a piezoelectric ceramic [36].

We mention here some of the considerations in determining the optimum wavelength or frequency of the source. Probably the most compelling reason for lengthening the wavelength is that most of the dimensions in the system scale accordingly. Increased dimensions facilitate fabrication of the passive components as well as their connection to the fiber loop. In addition, the power loss in the fiber due to Rayleigh scattering varies as the frequency to the fourth power and the Rayleigh backscatter noise has terms which vary as the amplitude, i.e., frequency squared, and the power of the scattered light. On the other hand the phase difference for a given rotation rate is reduced with the frequency though only to its first power and efficient detectors at long wavelengths may be difficult to make.

Aside from the above requirements and considerations other factors exist, but their importance is not yet fully understood. Frequency stability may be important for a stable scale factor as discussed later. The center frequency of the gain band of a He-Ne laser is stable and known to much better than the inertial navigation requirement of parts in 10^6 . The center frequency of the gain band of diode sources tends to drift with temperature by about a part in $10^3/^\circ\text{C}$ [89]. On the other hand, diode sources are small, highly efficient and can be produced at low cost. They also can be made to have a very wide-band output which may serve to reduce the tight tolerance on the polarizer [24] and decrease polarization-dependent scale factor instability [35], [90].

F. Single-Mode Fibers

The required characteristics of the single-mode fibers used in the fiber gyroscope have not been completely determined. These depend on factors which, themselves, are not fully specified. The fiber design determines the effect that a chang-

ing environment will have on the guided wave. The environment in which the system will operate depends on the application, and the requirements on the guided wave depend in part on the design of the system.

A major consideration is the polarization stability of the two interfering waves. In a reciprocal gyroscope, a change in the state of polarization can cause a change in the signal level as well as a phase difference error if the polarizer is not ideal (see Section II-D). Highly birefringent fibers have been designed specifically to stabilize the state of polarization of the guided wave [16]. These are likely to be very important in many systems. Alternatively, one gyroscope uses a wide-band source and a Lyot depolarizer to substantially reduce its sensitivity to the environmentally varied birefringence in standard fiber [35].

Another consideration is the stability of the phase difference between the two interfering waves. This is affected by time-varying temperature and pressure gradients across the fiber coil (see Section III-B). It has been demonstrated that the effect of the ambient pressure can be reduced by coating the fiber with a properly designed metallic jacket [91]. In addition, or as an alternative, the system can be made less sensitive to these gradients by wrapping the fiber coil symmetrically [28], [30].

G. Systems

The most sensitive systems reported to date have used, as a foundation, the reciprocal configuration of Section II-C in combination with the dynamic biasing technique described in Section II-E. The first of this type used bulk-optic beamsplitters and polarizers with a fiber-optic phase modulator and polarization controller, as shown schematically in Fig. 13(a) [14]. Considerable simplification occurred with the introduction of a fiber-optic directional coupler, in place of the beamsplitter and lenses, to split and combine the waves in the interferometer as shown in Fig. 13(b) [32]. The number of components was substantially reduced as were the difficulties of alignment and stabilization. To further this cause a fiber-optic polarizer along with a second directional coupler were introduced to form a gyroscope wherein the light is guided from source to detector (see Fig. 13(c)) [33]. A second polarization controller may be necessary to compensate for the birefringence in the section of fiber between the source and the polarizer. These systems, without modification, are limited by short-term noise on the order of $50^\circ/\text{h}$ due to Rayleigh scattering in the fiber of the coherent light from the helium-neon laser source [31], [32]. The most effective reduction of this noise came with the use of light with a relatively short coherence length such as that obtained from gain-guided semiconductor injection lasers [30], [32] or frequency-swept helium-neon lasers [36], as pointed out in Section III-C and IV-E. Short-term rms noise levels of about a degree per hour or less with integration time of 1 s^1 were

¹For comparison it is important to note that the rms value of white noise is proportional to the square root of the detection bandwidth or inversely proportional to the square root of the integration time. Dividing by the square root of the bandwidth we obtain a rotation-rate noise figure which, at least for white noise, is independent of bandwidth and is expressed in terms of degrees/ $\sqrt{\text{hour}}$. This figure is often used for comparison.

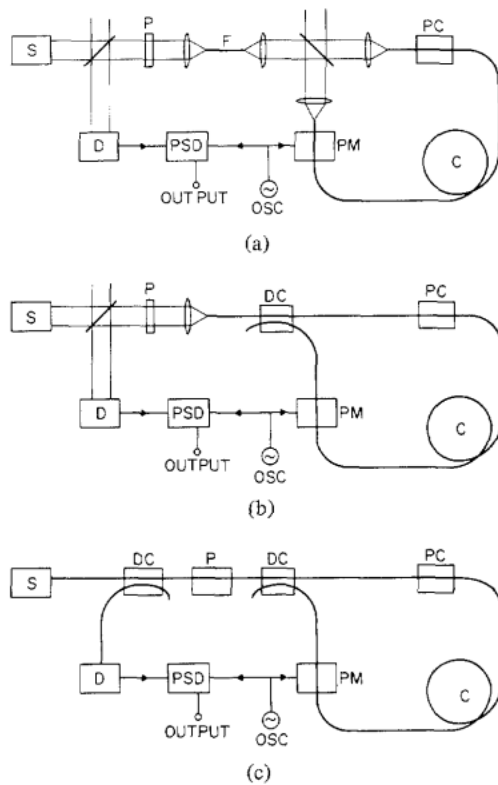


Fig. 13. Schematics of fiber gyroscopes with source *S*, polarizer *P*, filter *F*, polarization controller *PC*, phase modulator *PM*, oscillator *OSC*, phase sensitive detector (lock-in amplifier) *PSD*, detector *D*, and directional coupler *DC*.

then measured [30], [32], [36]. Further reductions of short-term noise to rms levels approaching 0.1 deg/h for the same 1-s integration time were obtained as the limitations became better understood. The reasons for these reductions are difficult to trace but they include giving proper attention to time-varying temperature and pressure gradients (see Section III-C), increasing the optical power incident on the detector to increase the signal with respect to the photon and electronic noise (see Section II-F), and the use of the proper modulation frequency to remove the residual coherent Rayleigh backscattering noise from the detection passband [37]. Further simplification was achieved by using highly birefringent polarization-preserving fibers, which obviates the need for polarization controllers [50]. Noise levels equivalent to 0.06 °/h rms with the 1-s integration time have been reported [37], [50]. These satisfy the long-range navigation short-term noise requirement. An example of this performance with an 8-s integration time is given in Fig. 14. Primary contributors at this noise level are photon, thermal electronic, and source-intensity noise within the passband of the detection system [37].

The long-term performance has had a similar history. A dramatic improvement from the rotation-rate error of tens of thousands of degrees per hour to about earth rate was obtained with the use of the reciprocal configuration (see Section II-C). Further improvement required consideration of the residual polarization-induced nonreciprocity due to a nonideal polarizer (see Section II-D) as well as the errors resulting from the Faraday and Kerr effects (see Sections III-D and III-E). In addition,

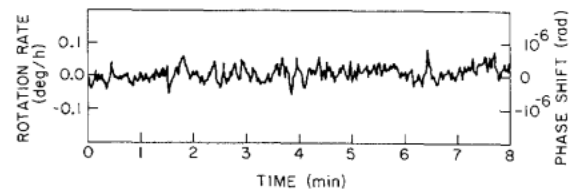


Fig. 14. Example of short-term performance with 8-s integration time.

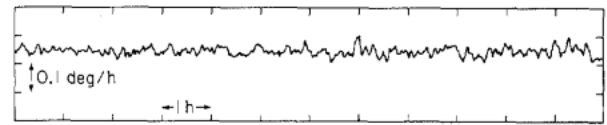


Fig. 15. Example of long-term performance.

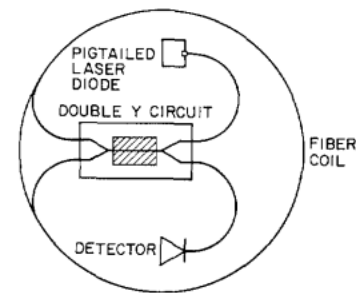


Fig. 16. Schematic of brass-board model of a fiber gyroscope using an integrated-optic circuit. The phase modulator which might be included in the chip is not shown.

unstable offsets, such as that due to improper filtering in the detection electronics or due to an imperfect phase modulator used for bias, required reduction. The latter is all but removed with the use of the proper bias modulation frequency. Accommodating these factors have led to signals whose short-term average value does not drift by more than about ± 0.01 °/h for periods of many hours [46], [50]. An example of this behavior is shown in Fig. 15.

Modifications to the basic design include the use of a Lyot depolarizer in the loop to reduce the scale-factor dependence on the environmentally sensitive fiber birefringence [37] and the use of Bragg cells in the loop to null the signal by frequency shifting the counterpropagating waves for closed-loop operation [36]. These changes are intended for more accurate measurements of nonzero rotation rates and, in particular, large rotation rates. We have not thus far considered rotation rates which differ significantly from zero and will discuss the problems associated with such measurements in Section IV-I when considering the scale factor.

A reciprocal fiber gyroscope with integrated-optic components was demonstrated by using a double-Y-configuration in a system as shown in Fig. 16 [57], [60]. A metal film was used on the center waveguide as a polarizer to effect the required single-mode filter. Instead of inducing phase modulation electrooptically on the chip, a fiber-optic phase modulator was used for simplicity. The circuit was fabricated on a LiNbO₃ substrate by Ti in-diffusion. An assembly of low-thermal-expansion ceramic and V-groove silicon blocks have been designed for coupling of fiber pigtailed to the circuit wave-

guide. Unfortunately, optical insertion loss through the planar optical circuit including the butt joints between the chip and fibers is excessive, as high as 20 dB for a single pass. As a result, the experimental sensitivity using a 500-m fiber wrapped in a coil of 18 cm diam was limited to $10^\circ/\sqrt{\text{h}}$. This demonstration, however, indicates the feasibility of a fiber gyroscope using integrated-optic components if the problems of the component quality are solved, and strong efforts are continuing in this direction.

H. Alternative Approaches

One alternative approach is a re-entrant fiber gyroscope [13], [92] which possesses some features of both the ring-laser gyroscope and the usual fiber gyroscope, and avoids some problems of both. Like the ring-laser gyroscope it recirculates the optical energy many times around a closed sensing loop which contains a gain medium, it has automatic built-in signal processing with output in the form of a frequency having the same value as the beat frequency from the ring-laser gyroscope, and it is an integrating-rate gyroscope having inherently linear response. It is like the usual fiber gyroscope in that it can be realized in all-solid-state form, contains an N-turn fiber sensing loop, uses an external fixed-frequency optical source, and is free of lock in. Unlike either, however, optical pulses of several microseconds duration, rather than CW light, are injected into the sensing loop from the external source.

Passive ring resonators in all-bulk-optic [4], [42], [93], all-integrated-optic [94], and all-fiber-optic [95], [96] forms have been made to sense rotation. Light from a coherent CW laser is divided and used to excite the resonator in the two opposite directions around the ring. Rotation changes the resonant condition for the two waves differently. The output for each direction is detected and the two are compared electronically. A frequency difference between the two waves at the input that aligns the resonances is proportional to the rotation rate much like the frequency difference of the two output waves from the ring-laser gyroscope.

Active fiber-ring resonators which make use of stimulated Brillouin scattering have been proposed for rotation sensing [97]. It has been argued that since the counterpropagating waves are amplified by different media, distinct counterpropagating acoustic waves, the problem of lock-in which occurs in other types of active resonators, will be absent [97]. Such a system has yet to be demonstrated and the claim for the absence of lock-in remains unsubstantiated.

Mach-Zender interferometers have also been used to sense rotation [98]. Like in the passive resonator, light from a laser is divided and passed through the Mach-Zender in opposite directions. The two outputs, one for each direction of propagation, vary differently with rotation. These are detected and a comparison is made electronically to determine the rotation rate. Each output is placed on an IF carrier by modulating the source frequency, and the phase difference between the electrical signals determines the rotation rate.

Finally, the reciprocity of a system using phase conjugation has been demonstrated [99]. The interferometer which was used bears resemblance to both Michelson and Sagnac interferometers with many of the properties of the latter. Recipro-

cal phase shifts were shown not to affect the output and convincing arguments were presented that it would be affected by nonreciprocal phase shifts.

V. FUTURE RESEARCH

A. Perspective

In this section, we group the research on the fiber gyroscope into two main categories. The first deals with the characteristics of the system, such as noise and instabilities, which are relatively independent of its rate of rotation, and the second encompasses the characteristics of the scale factor, i.e., the factor that relates the rotation rate to the output signal, which for the most part becomes increasingly important as the rotation rate is increased.

The vast majority of the work thus far has focused on the first category. This was accomplished by examining the system at rest in an inertial space where the effects of scale factor instabilities are minimized. As we have seen in the previous sections, many error sources in this category have been identified and methods to reduce their effects have been successfully implemented. As a result, it appears that the fiber gyroscope has become an accurate null-rotation-rate sensor. Further documentation in the areas of long-term stability and offsets is required for this to be confirmed. Research in this category continues to provide additional information about the error sources and alternative methods of dealing with them as well as additional system documentation.

In comparison to the low-rotation-rate studies, little progress has been made in the scale-factor category. The primary concerns here are the fractional instabilities of the scale factor, the dynamic range over which sensitive measurements can be made, and the possibilities for a digital output to allow the data to be processed. These issues as well as some system modifications designed to improve various scale-factor characteristics are discussed below. Before proceeding we note that changes in the system design must be approached somewhat cautiously, keeping in mind the constraints imposed by the known rotation-rate-independent noise. It will be of little value, in some cases, to improve scale-factor characteristics while sacrificing the achieved system sensitivity.

B. Open-Loop Operation

The output signal S of a fiber gyroscope is the result of the interference of two waves, and, when suitably biased as discussed in Section II-E, is given by

$$S = S_0 \sin(T_r \Omega)$$

where Ω is the rotation rate, S_0 is the fringe amplitude, and T_r is a constant with the units of time such that $T_r \Omega$ is the phase difference between the two interfering waves. This type of signal has a number of potential drawbacks depending upon the application. First, the output signal is a nonlinear function of the rotation rate with a dynamic range limited by the sinusoidal waveform. Assuming a measurement sensitivity of 10^{-7} rad we obtain a dynamic range on the order of 10^7 . Some applications require a dynamic range of 10^8 or more. Second, the output signal is in the form of an analog electrical

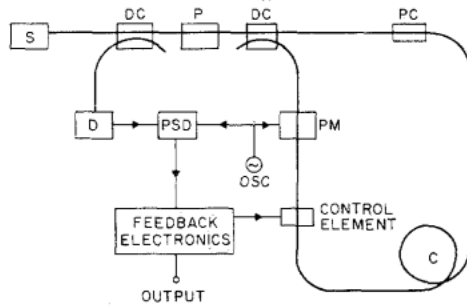


Fig. 17. Schematic of closed-loop fiber gyroscope where feedback electronics drive a control element in the interferometer which provides a phase shift to compensate for the Sagnac phase drift and null the signal from the phase-sensitive detector (PSD). The components are labeled consistent with those in Fig. 13.

current which, as a result of the enormous amount of information, is not easily digitized for processing. Thus this type of gyroscope is difficult to use in an inertial reference system where the data from three gyroscopes and three accelerometers must be simultaneously analyzed. Third, the stabilities of the quantities S_0 and T_r are questionable. The fringe amplitude S_0 varies with the source optical power, the optical power loss through the system, the states of polarization of the interfering waves, and the detection apparatus, to name a few. These variations can possibly be determined through independent measurements and removed from the signal. It is more difficult to visualize the removal of variations of T_r . Explicitly we write, according to (4), $T_r = 2\pi LD/\lambda c$, where L is the length of the fiber, D is the diameter of the fiber coil, c is the speed of light in vacuum, and λ is the mean source wavelength in vacuum. Of these quantities the wavelength is possibly the most difficult to stabilize to a high precision, at least if diode sources are employed (see Section IV-E).

C. Closed-Loop Operation

To circumvent some of the above difficulties, closed-loop approaches are being investigated. In a closed-loop gyroscope a feedback mechanism maintains the open-loop signal at zero by compensating for the Sagnac phase shift by introducing an equal and opposite phase shift within the sensing loop (Fig. 17). The measure of this added phase shift then reveals the rotation-rate information. A number of mechanisms are possible candidates. The characteristics of the one chosen will largely determine the characteristics of the scale factor. One approach which has been demonstrated employs frequency shifters located off center within the fiber loop, e.g., bulk-optic Bragg cells [36]. A frequency difference $\Delta\nu$ between the counterpropagating waves causes a difference of

$$\Delta\phi_{\Delta\nu} = 2\pi\Delta\nu \frac{nL}{c} \quad (24)$$

where n is the effective refractive index of the fiber, which is assumed to be constant over the small frequency range involved. By tuning the frequency difference, the phase difference is adjusted to be equal in magnitude and opposite in sign to the Sagnac phase difference as determined by nulling the open-loop signal. Using (4) we obtain the new relation

$$\Delta\nu = \frac{D}{\lambda L} \Omega \quad (25)$$

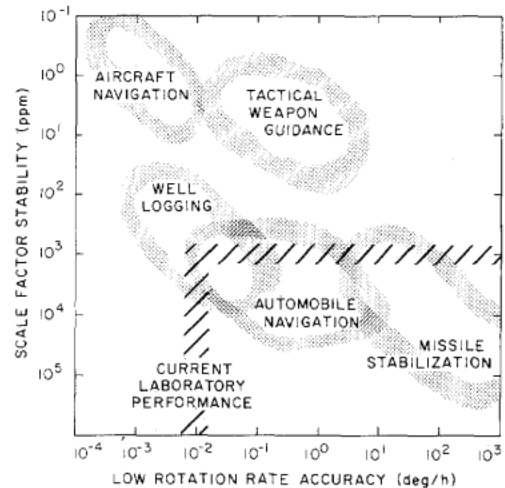


Fig. 18. Ranges of low-rotation-rate and scale-factor requirements for five classes of applications with an outline of current laboratory performance of state-of-the-art fiber gyroscopes.

It has been pointed out that this is the same result as that obtained for the ring laser and the passive resonator gyroscopes [36]. The limitations now arise in our ability to stabilize D , n , and λ , and in our ability to create and determine $\Delta\nu$. This type of rotation-rate signal has the potential for an extremely large dynamic range and, since it is in the form of a frequency difference, for an accurately digitized output. The major drawback to this approach is that adequate frequency shifters have not been demonstrated in guided-wave form, and the use of bulk-optic components such as Bragg cells in the fiber loop contribute certain size, stability, and alignment difficulties.

Another approach is one which makes use of the phase-modulation biasing technique. By applying a second phase modulation in the loop at twice the frequency of, and phase-matched to the biasing modulation the rotation-rate signal can be nulled [100]. The amplitude of the second-harmonic signal then provides the rotation rate information [100]. This has the advantage that it has been demonstrated using only fiber-optic components and it has the potential for the suppression of the source wavelength dependence if the scale factor is made linear. It has the disadvantage that accurate knowledge of the amplitude of the phase modulation is critical and techniques for obtaining this have yet to be demonstrated.

VI. APPLICATIONS

Continued improvement in the performance of fiber-optic gyroscopes has promoted interest in the applications of these systems. In order to address questions of suitability, we first consider two important performance criteria. They are low-rotation-rate accuracy, where the effects of variations of the scale factor are minimized, and scale-factor stability. One or the other of these two limit the accuracy about any given rotation rate. In Fig. 18 we have indicated the required range for these two criteria for five classes of applications. In addition, we have outlined the current laboratory performance of state-of-the-art systems. As a goal, sights have been set on satisfying the requirements for aircraft navigation. Such systems must operate accurately over a dynamic range of eight orders of magnitude, and, therefore, require not only good low-rotation-rate accuracy but also a very stable scale factor. The latter is discussed in the previous section and remains a problem to be

solved. Tactical-weapon navigation also requires a stable scale factor for the same reason. Because of a reduced time-of-flight for these systems, the rate-accuracy requirements are less stringent. In addition to satisfying the scale-factor requirement, the sensor must be able to operate while undergoing large accelerations and vibrations to be effective. Well logging is intended to determine the orientation of oil wells with respect to coordinates on the earth. This can be accomplished by measuring components on the earth's rotation vector which are on the order of $10^\circ/\text{h}$. The desired accuracy is between 0.01 and 1 percent in angle. These are achieved within the indicated ranges of sensitivity and scale-factor stability. The current laboratory performance of the fiber gyroscope extends into these ranges. The major problem then is to extend this performance to the environment of the well, where temperatures typically extend to 200°C . The device may also be required to withstand substantial shocks, though not during the measurement. Automobile navigation is to be used primarily to help dispatchers keep track of the positions of their vehicles, e.g., police cars, taxi cabs, etc. The most difficult criterion to overcome in this case appears to be the cost of this system. This may be limited to a few hundred U.S. dollars. Missiles are stabilized so that sensors are able to locate and inspect a target. Here the system must be able to withstand large vibrations and accelerations.

VII. CONCLUSION

We have seen that an optical fiber has been used to substantially increase the Sagnac phase difference due to rotation. This phase difference has then been measured accurately by making use of reciprocity and dynamic biasing technique and by removing the effect of environmentally induced dynamic variations in the system acting alone and together with Rayleigh backscatter, the Faraday effect, and the optical Kerr effect. We have reviewed the technology involved in making these systems in hybrid and all-guided forms. Performance levels have been given which demonstrates the accurate measurement capability of these systems for small rotation rates. The problem of extending this measurement to high rotation rates has been outlined.

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